



SCEGGS Darlinghurst

2008

**Preliminary Course
Semester 2 Examination**

Mathematics Extension 1

Outcomes Assessed: PE2 – PE6

Task Weighting: 40%

General Instructions

- Reading time – 5 minutes
- Working time – 1½ hours
- This paper has **five** questions
- Write using blue or black pen
- Answer all questions on the pad paper provided
- Write your Student Number at the top of each page
- Attempt **all** questions and show all necessary working
- **Start each question on a new page**
- Marks will be deducted for careless or badly arranged work
- Mathematical templates, geometrical equipment and scientific calculators may be used

Total marks – 60

- Attempt Questions 1 – 5

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Question 1 (12 marks)

- (a) The point P divides the interval AB joining $A(-2, -3)$ and $B(1, 2)$ externally in the ratio $3 : 2$. 2

Find the co-ordinates of P .

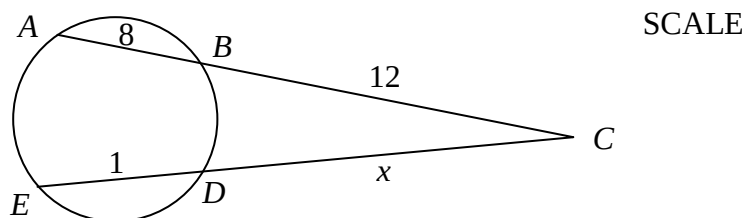
- (b) The equation $2x^3 - 4x - 7 = 0$ has roots α , β and γ .
Find the value of:

(i) $\alpha\beta\gamma$ 1

(ii) $\alpha\beta + \beta\gamma + \alpha\gamma$ 1

(iii) $\alpha^2 + \beta^2 + \gamma^2$ 2

- (c) 3



In the diagram ABC and EDC are straight lines.

$AB = 8\text{cm}$, $BC = 12\text{cm}$ and $DE = 1\text{cm}$

Find x giving reasons.

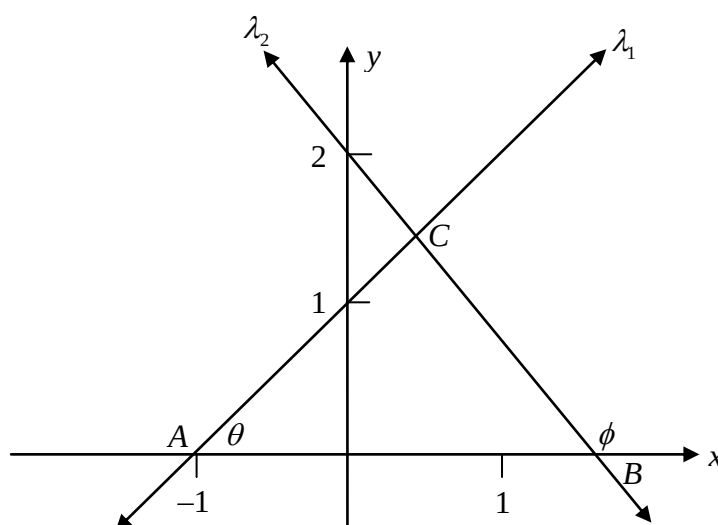
- (d) A polynomial is given by $P(x) = x^3 + ax^2 + bx + 6$. Find the values of a and b if $(x + 3)$ is a factor and if 12 is the remainder when $P(x)$ is divided by $(x + 1)$ 3

Question 2 (12 marks)

(a) (i) Express $\sqrt{3} \cos \theta + \sin \theta$ in the form $A \sin(\theta + \alpha)$ where $A > 0$. 2

(ii) Hence solve the equation $\sqrt{3} \cos \theta + \sin \theta = -\sqrt{3}$ for $0^\circ \leq \theta \leq 360^\circ$. 2

(b) The line λ_1 has the equation $x - y + 1 = 0$ and meets the x -axis at A . The line λ_2 has the equation $\sqrt{3}x + y - 2 = 0$ and meets the x -axis at B . λ_1 and λ_2 meet at C .



(i) Find the exact value for $\tan \angle ACB$ ($\angle ACB$ is acute) in its simplest form. 2

(ii) Find θ and ϕ and hence show $\angle ACB = 75^\circ$. 2

(iii) Hence find the exact value of $\tan 75^\circ$ 1

Question 2 continues on the next page

		Marks
Question 2 (continued)		
(c)	(i) How many words can be created from the letters of the word COONABARABRAN.	1
	(ii) What is the probability that a word chosen at random has all the "A"s together?	2

Question 3 (12 marks)

(a) Let $P(x) = (x - 2)(x - 1)^2(x + 2)^3$

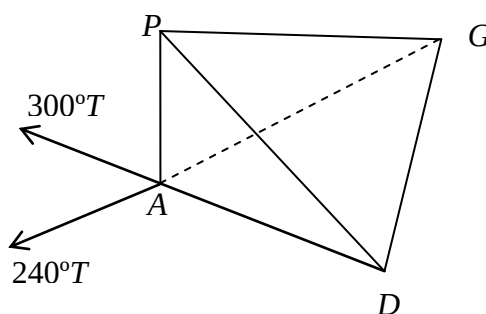
(i) Evaluate $P(0)$. 1

(ii) Sketch $y = P(x)$ labelling all important features 3

(b) (i) If there are 8 men and 6 women, how many committees of 5 people can be chosen? 1

(ii) If a committee is chosen by random find the probability that it would have a majority of men. 2

(c) The diagram below shows Donna standing at D on level ground, whilst Gemma is standing 2000m away at G on the same level ground. They both take the bearing and elevation of a place P at the same instant. Donna finds the bearing is $300^\circ T$ and the angle of elevation 25° , whilst Gemma finds the bearing to be $240^\circ T$ and the angle of elevation 17° .



(i) Copy the diagram onto your sheet, showing all the information given. 1

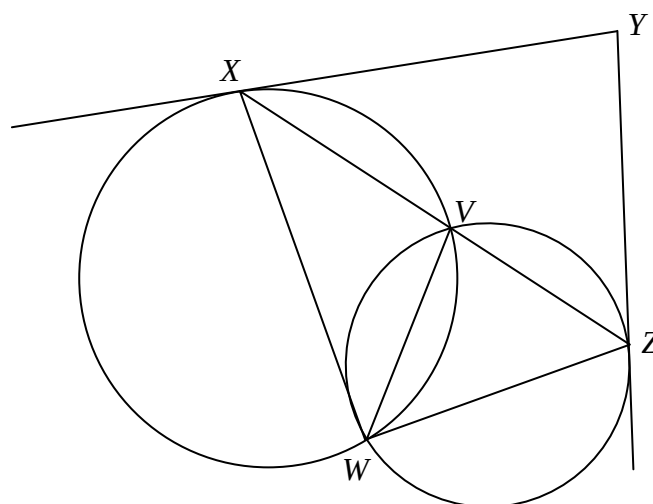
ii) Show that if the height PA of the plane is h metres then 3

$$h = \frac{2000}{\left(\tan^2 65^\circ + \tan^2 73^\circ - 2 \tan 65^\circ \tan 73^\circ \cos 60^\circ \right)^{\frac{1}{2}}}$$

(iii) Find h to 3 significant figures. 1

Question 4 (12 marks)

- (a) Todd and Meaghan go to the cinema with three other couples. They sit together as a group in a single row.
- (i) In how many ways can they be arranged? 1
- (ii) In how many ways can they sit so that each couple is together? 2
- (iii) Todd and Meaghan had an argument going into the cinema and decided they do not want to sit together. How many arrangements are possible if the other couples are still sitting with their partners? 2
- (b) Two circles intersect at V and W as shown. A line through V cuts the two circles at X and Z . The tangents at X and Z meet at Y . 3



Prove $XYZW$ is a cyclic quadrilateral.

Question 4 continues on the next page

Marks

Question 4 (continued)

- (c) (i) Sketch the graph of the polynomial $P(x) = x^3 - x^2 - 12x$ showing the intercepts on the x -axis. **2**
- (ii) Hence, solve the inequality $x - 1 \geq \frac{12}{x}$. **2**

Question 5 (12 marks)

(a) If $2^a + 3^b = 17$ and $2^{a+2} - 3^{b+1} = 5$ find the values of a and b . **2**

(b) Show that $\frac{\sin 5x}{\sin x} - \frac{\cos 5x}{\cos x} = 4 \cos 2x$ **3**

(c) Let $f(x) = \frac{x^2}{x^2 - 1}$

(i) For what values of x is $f(x)$ undefined **1**

(ii) Evaluate $\lim_{x \rightarrow \infty} \frac{x^2}{x^2 - 1}$ **1**

(iii) Find $f(0)$ and hence sketch the curve of $y = f(x)$ **3**

(iv) On the same axes sketch $y = x - 1$ **1**

(v) Hence find the number of solutions to $x^3 - 2x^2 - x + 1 = 0$ **1**
Explain your answer.

End of paper

Preliminary Course Extension / Semester 2 Examination 2008 - Solutions

Q1 a) A(-2, -3) B(1, 2)

$3: -2$

$x = \frac{3 \times 1 + (-2) \times (-2)}{3 + (-2)}$

$= \frac{3 + 4}{1}$
 $= 7$

$\therefore P(7, 12) \checkmark$

$y = \frac{3 \times 2 + (-2) \times (-3)}{3 + (-2)} \checkmark$

$= \frac{6 + 6}{1}$
 $= 12$

Several students have not learned the correct formula

b) i) $x \cdot y = -\frac{d}{a}$
 $= -\frac{(-7)}{2}$
 $= \frac{7}{2} \checkmark$

iii) $\alpha p + \beta y + \gamma x = \frac{c}{a}$
 $= \frac{-4}{2}$
 $= -2 \checkmark$

iii) $x^2 + y^2 + z^2 = (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \beta\gamma + \gamma\alpha) \checkmark$
 $= 0^2 - 2 \times -2$
 $= 4 \checkmark$

c) $AC \times BC = EC \times DC$ (product of the intercepts = length of secant through a point outside) $\checkmark$

$20 \times 12 = x(x+1)$

$x^2 + x - 240 = 0 \checkmark$

$(x-15)(x+16) = 0$

$x = 15 \quad \text{as } x > 0 \checkmark$

Comm - 3

There is no excuse for not knowing these formula
Be careful with coefficients
 $P(x) = 2x^2 - 4x - 7 \Rightarrow b = 0$

Only a few students could recall this property

d) $P(-3) = 0 \quad P(-1) = 12$

$\therefore (-3)^3 + a(-3)^2 + b(-3) + 6 = 0 \checkmark$

$-27 + 9a - 3b + 6 = 0$

$9a - 3b = 21$

$(-1)^3 + a(-1)^2 + b(-1) + 6 = 12 \checkmark$

$-1 + a - b + 6 = 12$

$a - b = 7$

$\therefore 9a - 3b = 21 \dots \textcircled{1}$

$a - b = 7 \dots \textcircled{2} \times 3$

$3a - 3b = 21 \dots \textcircled{3}$

$\textcircled{1} - \textcircled{3} \quad 6a = 0$

$a = 0$

using $\textcircled{2} \quad b = -7 \checkmark$

Recs - 3

Q2 a) i) $\sqrt{3} \cos \theta + \sin \theta = A \sin(\theta + \alpha)$

$= A \sin \theta \cos \alpha + A \cos \theta \sin \alpha$

$\therefore A \cos \alpha = 1$

$A \sin \alpha = \sqrt{3}$

$A^2 \sin^2 \alpha + A^2 \cos^2 \alpha = (\sqrt{3})^2 + 1^2$

$A^2 = 3 + 1$

$= 4$

$A = 2 \quad \checkmark \quad A > 0$

$\frac{A \sin \alpha}{A \cos \alpha} = \frac{\sqrt{3}}{1}$

$\tan \alpha = \sqrt{3}$

$\alpha = 60^\circ \checkmark$

$\therefore \sqrt{3} \cos \theta + \sin \theta = 2 \sin(\theta + 60^\circ)$

ii) $2 \sin(\theta + 60^\circ) = -\sqrt{3}$

$\sin(\theta + 60^\circ) = -\frac{\sqrt{3}}{2}$

$\theta + 60^\circ$ lies in the 3rd & 4th quadrant

$\theta + 60^\circ = 240^\circ \quad \theta + 60^\circ = 300^\circ$

$\theta = 180^\circ \quad \text{or } \theta = 240^\circ \checkmark$

Comm - 2

Some students confused the concepts of factor and remainder

Done very well.
Just be careful with the auxiliary angle
I saw 30° a few times!

Need to practise solving trig. equations
The quadrant work was poor.

b) i) $m_1 = 1$ $m_2 = -\sqrt{3}$
 $\tan \angle ACB = \left| \frac{1 - (-\sqrt{3})}{1 + 1 \cdot (-\sqrt{3})} \right|$ ✓
 $= \left| \frac{1 + \sqrt{3}}{1 - \sqrt{3}} \right|$
 $= \frac{\sqrt{3} + 1}{\sqrt{3} - 1}$ ✓

ii) $\tan \theta = 1$ $\tan \phi = -\sqrt{3}$
 $\theta = 45^\circ$ $\phi = 120^\circ$ ✓
 $\phi = \angle ACB + \theta$ (exterior angle equals sum of two opposite interior angles) ✓
 $120^\circ = \angle ACB + 45^\circ$
 $\angle ACB = 75^\circ$

iii) $\tan 75^\circ = \frac{\sqrt{3} + 1}{\sqrt{3} - 1}$ ✓ Recs - 5

c) i) $\frac{13!}{2!4!2!2!2!} = 16216200$ ✓ Comm-1

ii) No of words with A's together = $\frac{10!}{2!2!2!2!} = 226800$ ✓

$\therefore P(\text{A's together}) = \frac{226800}{16216200}$

$= \frac{2}{143}$ ✓

Recs - 2

Q3 a) i) $P(x) = (x-2)(x-1)^2(x+2)^3$
 $= -16$ ✓

ii) $y = -x^2 + x = 0$

$y = -16$

$x = 2$ $x = 1$ $x = -2$

multiplicity 1 2 3

First line was done very well but a majority of students didn't realise the impact of the $1/1$ sign.

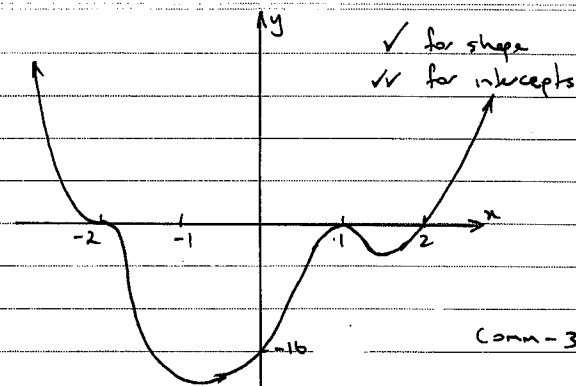
Again done well but many students left out the reasons (No penalty)

Done well

Many students made an addition error

i.e. $\frac{9!}{2!2!2!2!}$ or forgot

to divide by the 4 '2!'s

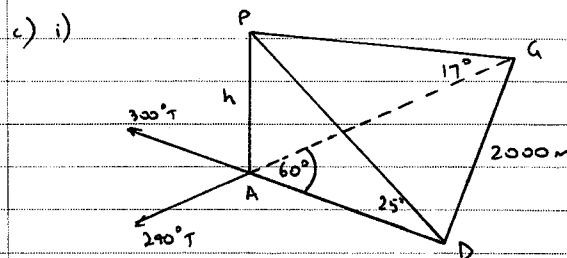


Students who used calculus were generally less successful

b) i) No of committees = ${}^{14}C_5 = 2002$ ✓

ii) No of committees with majority of men
 $= {}^8C_3 \times {}^6C_2 + {}^8C_4 \times {}^6C_1 + {}^8C_5 \times {}^6C_0$ ✓
 $= 840 + 420 + 56$
 $= 1316$

$\therefore P(\text{majority of men}) = \frac{1316}{2002} = \frac{94}{143}$ ✓ Recs - 3



Students needed to be convincing. Some students clearly "fudge" from the answer.

- Angles must be clearly identified - 3 letters

- Only a few students were able to show how 60° was calculated

- More supporting work are required.

ii) $\tan 65^\circ = \frac{AD}{h}$ $\tan 73^\circ = \frac{AG}{h}$
 $AD = h \tan 65^\circ$ $AG = h \tan 73^\circ$ ✓
 cosine rule: $DG^2 = AD^2 + AG^2 - 2AD \times AG \times \cos 60^\circ$
 $2000^2 = h^2 \tan^2 65^\circ + h^2 \tan^2 73^\circ$
 $- 2h^2 \tan 65^\circ \tan 73^\circ \cos 60^\circ$

$= h^2 (\tan^2 65^\circ + \tan^2 73^\circ - 2 \tan 65^\circ \tan 73^\circ \cos 60^\circ)$
 $h^2 = \frac{2000^2}{\tan^2 65^\circ + \tan^2 73^\circ - 2 \tan 65^\circ \tan 73^\circ \cos 60^\circ}$ ✓

$$h = \frac{2000}{\sqrt{\tan^2 65^\circ + \tan^2 73^\circ - 2 \tan 65^\circ \tan 73^\circ \cos 60^\circ}} \quad \text{Recs-3}$$

$$\text{iii) } h = 695 \text{ (to 3 sig. figs.)} \quad \checkmark$$

$$\text{Q4 a) i) No. of arrangements} = 8! = 40320 \quad \checkmark$$

$$\text{ii) No. of arrangements} = 4! \times 2! \times 2! \times 2! \times 2! = 384 \quad \checkmark$$

$$\text{iii) } \begin{array}{cccc} T & M & M & M \\ & T & M & M \\ & & T & M \end{array} \quad \left. \begin{array}{c} \\ \\ \end{array} \right\} 6 \text{ ways} \quad \checkmark$$

$$\therefore \text{No. of arrangements} = 6 \times 2 \times 3! \times 2! \times 2! \times 2! = 576 \quad \text{Recs-5}$$

$$\text{b) let } \angle YXZ = \alpha \text{ and } \angle YZX = \beta$$

$$\therefore \angle XYZ = 180 - (\alpha + \beta) \quad (\text{angle sum of a triangle is } 180^\circ) \quad \checkmark$$

$$\angle ZWV = \beta \quad (\text{angle at tangent equals angle in the alternate segment}) \quad \checkmark$$

$$\angle XWV = \alpha \quad (\text{ " " " }) \quad \checkmark$$

$$\angle XWZ = \angle XWV + \angle ZWV = \alpha + \beta$$

$$\therefore \angle XWZ + \angle XYZ = 180$$

$$\therefore XYZW \text{ is a cyclic quadrilateral as opposite angles are supplementary.} \quad \checkmark \quad \text{Conn-3}$$

$$\begin{aligned} \text{c) i) } P(x) &= x^3 - x^2 - 12x \\ &= x(x^2 - x - 12) \\ &= x(x-4)(x+3) \end{aligned}$$

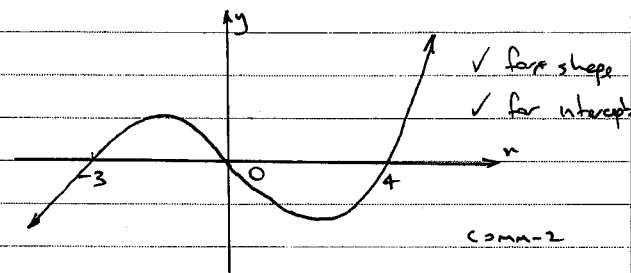
This was done well.

parts ii) & iii) were done poorly but they were tricky. Have a look at the solution and ask questions

TANGENTS FROM DIFFERENT CIRCLES ARE NOT EQUAL IN LENGTH !!!

Everyone except 2 or 3 people thought they did.

$$\begin{aligned} \therefore y-nt: x=0 \quad P(0) &= 0 \\ x-nt: y=0 \quad P(n) &= 0 \\ x(x-4)(x+3) &= 0 \\ x=0 \quad x=-3 \quad x=4 \end{aligned}$$



$$\begin{aligned} \text{ii) } x^3(x-1) &\geq \frac{12}{x} \times x^2 \\ x^3 - x^2 &\geq 12x \\ x^3 - x^2 - 12x &\geq 0 \quad \checkmark \\ \therefore \text{from the graph } -3 \leq x < 0 \text{ and } x > 4 \quad \checkmark \\ &\quad \text{Recs-2} \end{aligned}$$

$$\text{Q5 a) let } m = 2^a \text{ and } n = 3^b$$

$$\therefore 2^a + 3^b = 17 \quad 2^{a+2} - 3^{b+1} = 5$$

$$m+n=17 \quad 2^2 \times 2^a - 3 \times 3^b = 5$$

$$4m - 3n = 5$$

$$\therefore 4m - 3n = 5 \dots \textcircled{1}$$

$$m+n=17 \dots \textcircled{2} \times 3 \quad \checkmark$$

$$3m+3n=51 \dots \textcircled{3}$$

$$\textcircled{1} + \textcircled{3}$$

$$7m = 56$$

$$m = 8$$

$$\text{from } \textcircled{2} \quad n = 9$$

$$\therefore 2^a = 8 \quad 3^b = 9$$

$$a = 3 \quad b = 2 \quad \checkmark \quad \text{Recs-2}$$

Done very well. A few people graphed it the wrong way.

Many students missed the link between i) & ii) because they ended up with $x^2 - x - 12 \geq 0$. You have to multiply through by x^2 not x to keep $\geq$.

Only a few students were successful in this question.

Alternative solutions are possible but only a couple of students were able to correctly find 'a' and 'b'.

A couple of "lucky" students "chanced" upon the correct answer by trial and error.

$$b) \text{ LHS} = \frac{\sin 5x}{\sin x} - \frac{\cos 5x}{\cos x}$$

$$= \frac{\sin 5x \cos x - \cos 5x \sin x}{\sin x \cos x} \quad \checkmark$$

$$= \frac{\sin(5x - x)}{\sin x \cos x} \quad \checkmark$$

$$= \frac{\sin 4x}{\frac{1}{2} \sin 2x}$$

$$= \frac{2 \sin 2x \cos 2x}{\frac{1}{2} \sin 2x} \quad \checkmark$$

$$= 4 \cos 2x$$

Reas-3

$$c) i) x = \pm 1 \quad \checkmark$$

$$ii) \lim_{x \rightarrow \infty} \frac{x^2}{x^2 - 1}$$

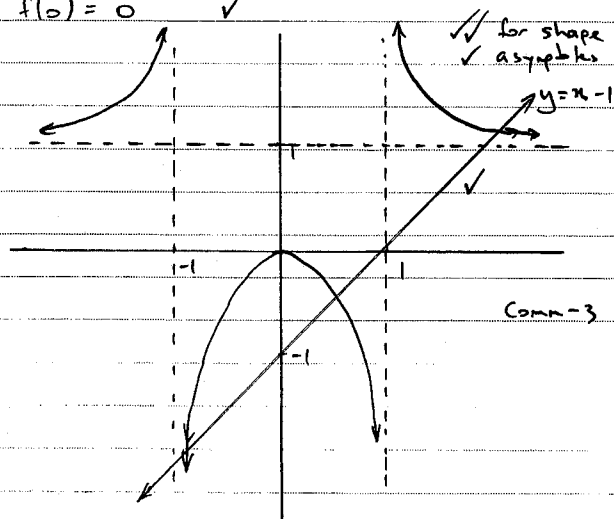
$$= \lim_{x \rightarrow \infty} \frac{\frac{1}{x^2}}{\frac{1}{x^2} - \frac{1}{x^2}}$$

$$= \lim_{x \rightarrow \infty} \frac{1}{1 - \frac{1}{x^2}}$$

$$= 1 \quad \checkmark$$

$$iii) f(0) = 0 \quad \checkmark$$

iv)



Well done by most students

Comm-3

v) Solve simultaneously

$$y = \frac{x^2}{x^2 - 1} \dots \textcircled{1} \quad y = x - 1 \dots \textcircled{2}$$

$$\textcircled{1} = \textcircled{2}$$

$$\frac{x^2}{x^2 - 1} = x - 1$$

$$x^2 = (x - 1)(x^2 - 1)$$

$$x^2 = x^3 - x - x^2 + 1$$

$$0 = x^3 - 2x^2 - x + 1$$

$\therefore$ point of intersection of $y = \frac{x^2}{x^2 - 1}$ and $y = x - 1$
are the solutions to $x^3 - 2x^2 - x + 1 = 0$

$\therefore$ 3 solutions. $\checkmark$

Reas-1

A clear statement of the reason was required to obtain this mark.